Athena Elvey

646-832-5801 | athenaelvey@gmail.com | linkedin.com/in/athenaelvey | github.com/athenaelvey

Education

Rensselaer Polytechnic Institute Troy, NY

B.S. Computer Science & Cognitive Science *Expected Graduation: Dec. 2028*

- **Honors:** AP Scholar with Distinction, Garnet D. Baltimore Scholarship, National African-American Recognition Program
- **Relevant Coursework:** Foundations of Computer Science, Data Structures, Linear Algebra, Cognitive Neuroscience, Computer Architecture and Operating Systems

Technical Skills

Languages: Java, Python, SQL, C++, C, JavaScript, HTML/CSS, Rust
Frameworks & Tools: React, Vite, React Router, Tailwind CSS, FastAPI, REST APIs, SQLAlchemy, Alembic, Pydantic, PostgreSQL, Docker, pytest, Git
Data & ML: Pandas, NumPy, scikit-learn, TensorFlow/Keras, Altair, Plotly, Matplotlib
Concepts: Data Structures, Full-Stack Development, Database Design, Data Visualization, Regression Modeling

Projects

Chronova | *React, Vite, FastAPI, PostgreSQL, SQLAlchemy, Docker, Three.js, Plotly, pytest* Chronova

- Built and deployed a full-stack explorer for the ATNF Pulsar Catalogue using a React/Vite frontend, a FastAPI REST API with Pydantic v2 validation, and a Dockerized PostgreSQL database with Alembic migrations
- Enabled flexible search by building a query builder with AND/OR combinators that validates each condition against its field's allowed operators before converting it to SQLAlchemy expressions
- Kept the dataset current without manual updates by automating ingestion with psrqpy and a weekly GitHub Actions workflow that refreshes the production PostgreSQL database
- Made pulsar physics easier to explore with an interactive Plotly period vs. period-derivative diagram, derived metrics for characteristic age and magnetic field strength, and Three.js 3D views of each pulsar

Daily Cognitive Load Tracker | *Python, Streamlit, Pandas, Altair, NumPy* DCLT

- Built and deployed a live wellness tracker for daily stress, energy, and focus, adding streak tracking and pattern-based alerts to encourage consistent logging
- Showed users their trends in real time by fitting NumPy linear regression lines to each metric, with CSV persistence
- Made patterns easy to read by building interactive Altair visualizations of focus, stress, and energy over time

Hospital Length-of-Stay Prediction Model | *Python, Pandas, scikit-learn, TensorFlow/Keras*

- Identified how medical equipment availability relates to patient length of stay by analyzing healthcare infrastructure data with a team, and compared linear regression, polynomial regression, and a tuned TensorFlow/Keras neural network to find the best predictor

Experience

Creative Mortgage Solutions Sep. 2026 - Present

Operations Support Intern

- Kept lender and client databases accurate, categorized, and accessible by maintaining and updating records while tracking outstanding requirements to move files toward closing
- Helped the operations team answer lenders and clients faster by organizing documentation, tracking open items, and flagging gaps
- Coordinated workflows across internal systems and Microsoft Teams to keep task status visible to the team

America Reads, America Counts, Tutor Time & STEP Sep. 2025 - Present

Tutor *Troy, NY*

- Improved test scores for 20+ grade school and high school students by tutoring them in literacy, math, and computer science

Leadership and Organizations

National Society of Black Engineers, RPI Chapter Sep. 2026 - Present

Communications Committee *Troy, NY*

- Informed members and the RPI community about NSBE events, programs, and opportunities by coordinating and sharing information

SEO Career Jul. 2026 - Present

SEO EDGE Participant *New York, NY*

- Built a foundation in core business concepts including strategy, finance, and professional communication through targeted coaching and structured training

RPI Undergraduate Council / Class of 2029 Council Oct. 2025 - Mar. 2026

Class of 2029 Vice President *Troy, NY*

- Represented the interests of 1,600 students by contributing to freshman well-being initiatives